

Characterizing Fiber Scaffolds in Vacuum Conditions

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Abstract

This paper reports a systematic evaluation of porous lattice alloys. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses.

1. Introduction

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The reduced model retains most of the explanatory power at a fraction of the computational cost. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Calibration drift over the trial period remained below the threshold set out in the protocol. Longer observation windows narrow the confidence interval but do not shift the central estimate.

Taken together, these findings argue for treating the two regimes separately in future analyses. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

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2. Related Work

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3. Materials and Methods

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4. Experimental Setup

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